

TRANSMISSION LINE CAPACITANCELesson Summary:

1. Electric field and Voltage Calculation
2. Transmission line Capacitance for:
 - a. Single phase line (solid conductor)
 - b. Three phase line with equal spacing.
 - c. 3 ϕ line bundled conductor & Unequal spacing.

Steps for Calculating Capacitance:

1. Gauss's Law $\rightarrow$ Electric field strength E .
2. Voltage b/w conductors V
3. Capacitance $C = q/V$.

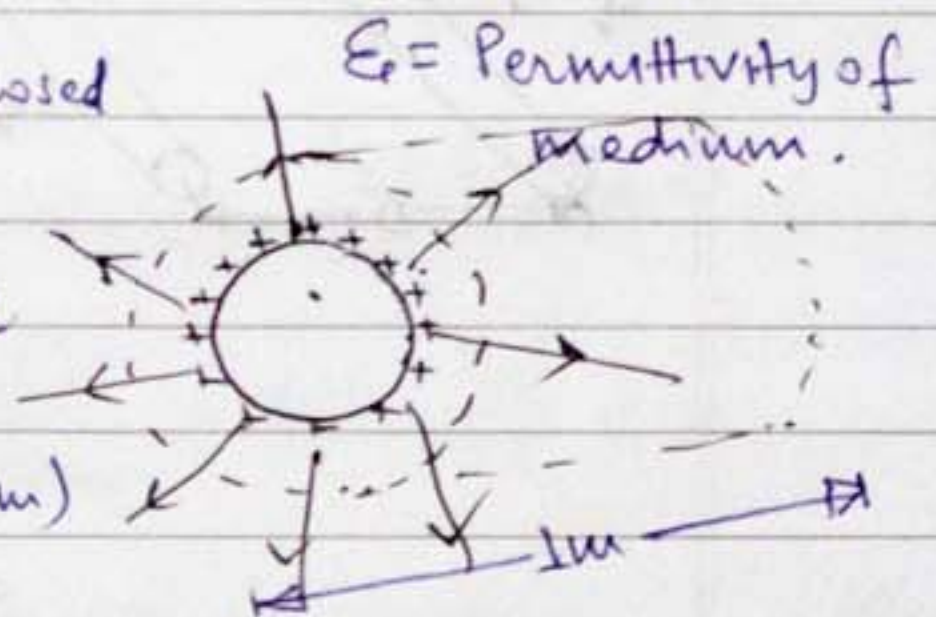
Gauss's Law:

Total electric flux leaving a closed surface = Q_{enclosed} .

$$\oint \mathbf{D} \cdot d\mathbf{s} = \oint \epsilon_0 \mathbf{E} \cdot d\mathbf{s} = Q_{\text{enclosed}}$$

Electric field line are radial to the conductor. There is no tangential field

$$\epsilon E \cdot \underbrace{(2\pi x)(1m)}_{\text{closed surface area}} = Q_{\text{enclosed}}(1m)$$



$$E = \frac{q}{2\pi \epsilon x} \text{ V/m.}$$

$$V_{12} = \int_{D_1}^{D_2} E_x dx = \frac{q}{2\pi \epsilon} \ln \frac{D_2}{D_1}$$

$$\epsilon = \epsilon_0 \epsilon_r.$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$C = \frac{q}{V} = \frac{2\pi \epsilon}{\ln D_2/D_1}$$

Multi-conductor system

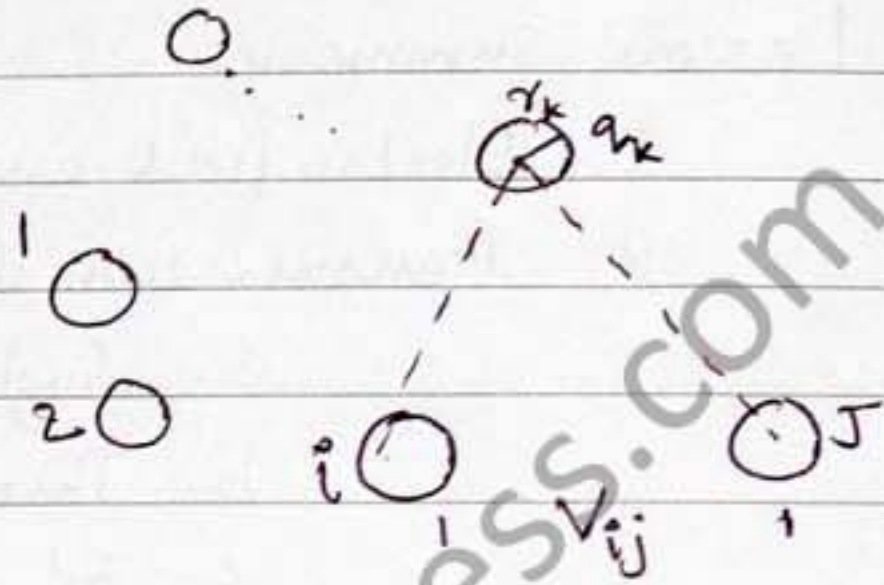
Conductor k has radius r_k and charge q_k .

Now,

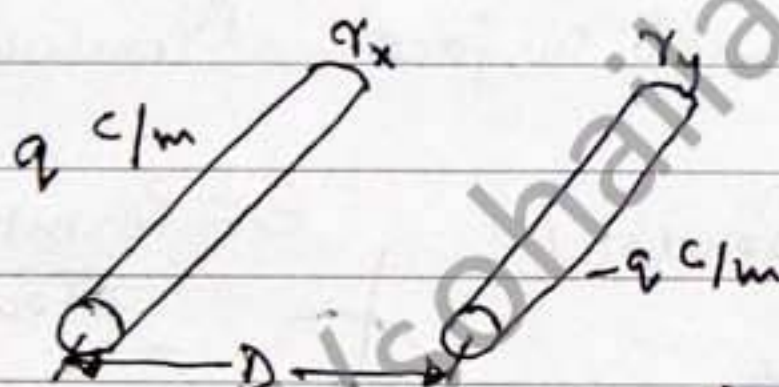
Voltage b/w conductor i & j due to charge q_k on conductor k is

$$V_{ijk} = \frac{q_k}{2\pi\epsilon} \ln \frac{D_{jk}}{D_{ik}}$$

$$\therefore V_{ij} = \sum_{k=1}^n \frac{q_k}{2\pi\epsilon} \ln \frac{D_{jk}}{D_{ik}}$$



2 (a) SINGLE PHASE LINE (Solid conductor)



$$V_{xy} = \frac{1}{2\pi\epsilon} \left[q \ln \frac{D_{yx}}{D_{xx}} - q \ln \frac{D_{yy}}{D_{xy}} \right]$$

$$V_{xy} = \frac{q}{2\pi\epsilon} \ln \frac{D_{yx} D_{xy}}{D_{xx} D_{yy}}$$

$$V_{xy} = \frac{q}{2\pi\epsilon} \ln \frac{D^2}{r_x r_y} = \frac{q}{\pi\epsilon} \ln \frac{D}{\sqrt{r_x r_y}}$$

$$C_{xy} = \frac{\pi\epsilon}{\ln \left(\frac{D}{\sqrt{r_x r_y}} \right)} \text{ F/m (line to line)}$$

If radius of two conductor are equal then.

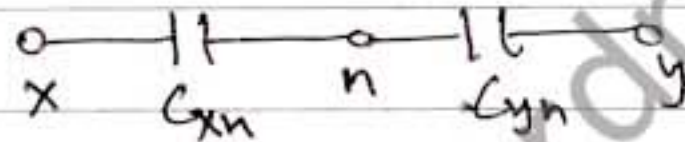
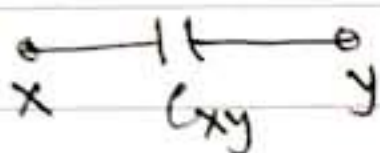
$$C_{xy} = \frac{\pi\epsilon}{\ln D/r} \text{ f/m (line to line)}$$

→ At the centre of x & y , potential will be zero.

$$\therefore V_{xn} = V_{yn} = V_{xy}/2.$$

$$C_n = C_{xn} = C_{yn} = \frac{q}{V_{xn}} = 2 C_{xy}.$$

$$C_n = \frac{2\pi\epsilon}{\ln(D/r)} \quad \text{F/m (line to Neutral)}$$



$$C_{xn} = C_{yn} = 2 C_{xy}.$$

(b) THREE PHASE LINE WITH EQUILATERAL SPACING

$$V_{ab} = \frac{1}{2\pi\epsilon} \left[q_a \ln \frac{D_{ba}}{D_{aa}} + q_b \ln \frac{D_{bb}}{D_{ab}} + q_c \ln \frac{D_{bc}}{D_{ac}} \right]$$

$$= \frac{1}{2\pi\epsilon} \left[q_a \ln \frac{D}{r} + q_b \ln \frac{r}{D} + q_c \ln \frac{D}{r} \right]$$

$$= \frac{1}{2\pi\epsilon} \left[q_a \ln \frac{D}{r} + q_b \ln \frac{r}{D} \right]$$

$$q_a + q_b + q_c = 0$$

$$V_{ac} = \frac{1}{2\pi\epsilon} \left[q_a \ln \frac{D_{ca}}{D_{aa}} + q_b \ln \frac{D_{cb}}{D_{ab}} + q_c \ln \frac{D_{cc}}{D_{ac}} \right]$$

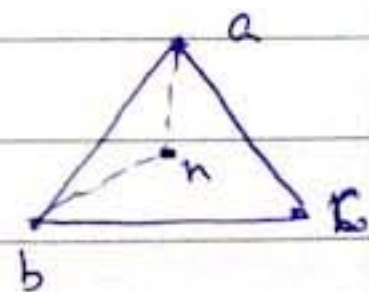
$$= \frac{1}{2\pi\epsilon} \left[q_a \ln \frac{D}{r} + q_c \ln \frac{r}{D} \right]$$

we know,

$$V_{ab} = \sqrt{3} V_{an} \angle +30^\circ$$

$$= \sqrt{3} V_{an} \left[\frac{\sqrt{3}}{2} + j\frac{1}{2} \right]$$

$$V_{ac} = -V_{ca} = \sqrt{3} V_{an} \left[\frac{\sqrt{3}}{2} - j\frac{1}{2} \right]$$



$$\therefore V_{ab} + V_{ac} = 3 V_{an}$$

$$V_{an} = \frac{1}{3} \left(\frac{1}{2\pi\epsilon} \right) \left[2q_a \ln \frac{D}{r} + (q_b + q_c) \ln \frac{r}{D} \right]$$

Since $q_a + q_b + q_c = 0$

$$\therefore V_{an} = \frac{1}{2\pi\epsilon} q_a \ln \frac{D}{r}$$

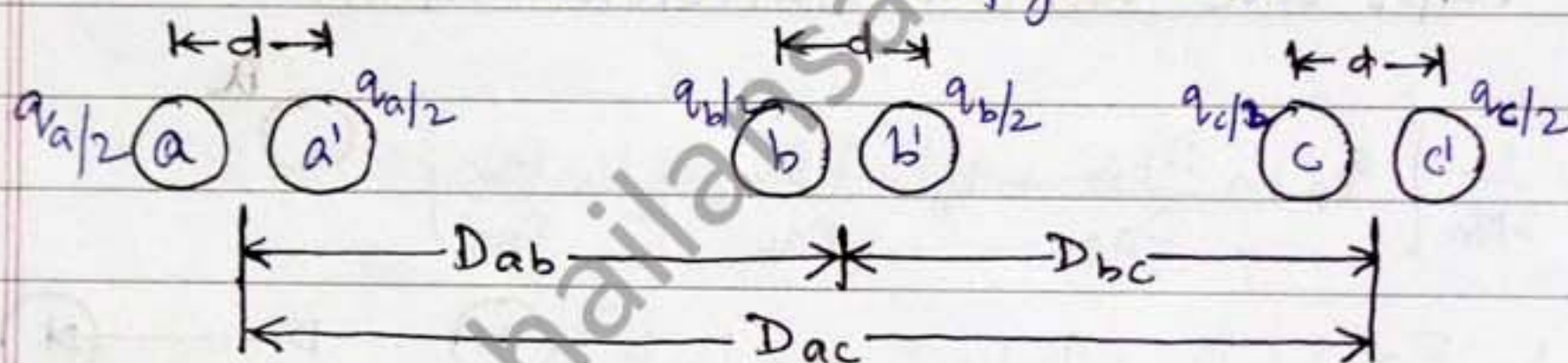
$$C_{an} = \frac{2\pi\epsilon}{\ln \frac{D}{r}} \text{ F/m line to Neutral}$$

In general,

$$C_{an} = \frac{2\pi\epsilon}{\ln \frac{D_{eq}}{r}} \text{ F/m}$$

Where, $D_{eq} = \sqrt[3]{D_{ab} D_{bc} D_{ac}}$

(C) 3 ϕ Bundled conductor lines with unequal spacing.
(Horizontal configuration).



$$V_{ab} = \frac{1}{2\pi\epsilon} \left[\frac{q_a}{2} \ln \frac{D_{ba}}{D_{aa}} + \frac{q_a}{2} \ln \frac{D_{ba'}}{D_{aa'}} + \frac{q_b}{2} \ln \frac{D_{bb}}{D_{ab}} + \frac{q_b}{2} \ln \frac{D_{bb'}}{D_{ab'}} \right. \\ \left. + \frac{q_c}{2} \ln \frac{D_{bc}}{D_{ac}} + \frac{q_c}{2} \ln \frac{D_{bc'}}{D_{ac'}} \right]$$

$$= \frac{1}{2\pi\epsilon} \left[\frac{q_a}{2} \left(\ln \frac{D_{ab}}{r} + \ln \frac{D_{ab}}{d} \right) + \frac{q_b}{2} \left(\ln \frac{r}{D_{ab}} + \ln \frac{d}{D_{ab}} \right) \right. \\ \left. + \frac{q_c}{2} \left(\ln \frac{D_{bc}}{D_{ac}} + \ln \frac{D_{bc'}}{D_{ac'}} \right) \right]$$

$$= \frac{1}{2\pi\epsilon} \left[q_a \ln \frac{D_{ab}}{\sqrt{rd}} + q_b \ln \frac{\sqrt{rd}}{D_{ab}} + q_c \ln \frac{D_{bc}}{D_{ac}} \right]$$

$\therefore$

$$C_{an} = \frac{2\pi\epsilon}{\ln(D_{eq}/D_s)}$$

$D_s = \sqrt{rd}$ for 2 conductor Bundle

$D_s = \sqrt[3]{rd^2}$ " 3 " "

$D_s = 1.091 \sqrt[4]{rd^3}$ " 4 " "

Numerical:

A three phase 400kV, 50Hz, 350km overhead transmission line has flat horizontal spacing with three identical conductors. The conductors have an outside diameter of 3.28 cm with 12m b/w adjacent conductor. Determine Capacitive reactance line to neutral in ohm/m/phase and capacitive reactance for the line in ohm/phase.

For fully transposed 3 ϕ line, we have line voltage given by

$$V_{ab} = \frac{1}{2\pi\epsilon} \left[q_a \ln \frac{D_{eq}}{r} + q_b \ln \frac{r}{D_{eq}} \right]$$

$$D_{eq} = (D_{12} D_{23} D_{31})^{1/3} = (12 \times 12 \times 24)^{1/3} = 15.119m$$

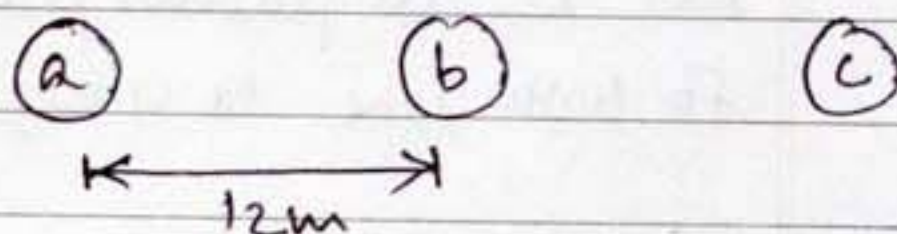
Similarly,

$$V_{ac} = \frac{1}{2\pi\epsilon} \left(q_a \ln \frac{D_{eq}}{r} + q_c \ln \frac{r}{D_{eq}} \right)$$

Now,

$$V_{ab} + V_{ac} = 3V_{an} \text{ \&}$$

$$q_b + q_c = -q_a$$



$$V_{an} = \frac{q_a}{2\pi\epsilon} \left(\ln \frac{D_{eq}}{r} \right)$$

$$C_{an} = \frac{2\pi\epsilon}{\ln \frac{D_{eq}}{r}} = 8.163 \times 10^{-6} \mu F/m.$$

$$Y_n = 2\pi f C_n = 2.565 \times 10^{-9} \text{ S/m/phase}$$

Length of line = 350km

$$\therefore Y_n = 8.978 \times 10^{-4} \text{ S/phase.}$$

$$X_n = \frac{1}{Y_n} = 1.1138 \times 10^3 \Omega / \text{phase.}$$

Lectures 06 are not in GATE syllabus.